File Permissions

Q. Write a shell script to modify file permissions and ownership using chmod and chown. Include user input for the filename and desired permissions, and display the new file permissions.

Code:

```
  GNU nano 7.2                                                    perm-own.sh
#!/bin/bash

echo "Enter file name:"
read file

echo "Enter permission (example 755):"
read perm

chmod $perm $file

echo "Enter new owner name:"
read owner

sudo chown $owner $file

echo "New file permissions are:"
ls -l $file
```

Output:

```
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ whoami
rvu
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ nano perm-own.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ chmod +x perm-own.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ ./perm-own.sh
Enter file name:
test.txt
Enter permission (example 755):
755
Enter new owner name:
rvu
New file permissions are:
-rwxr-xr-x 1 rvu rvu 0 Mar 10 14:18 test.txt
```

Q.Write a shell script to modify file permissions and ownership using chmod and chown. Include user input for the filename and desired permissions, and display the new file permissions.

Code:

```
  GNU nano 7.2                                                    mod-perm.sh
#!/bin/bash

# --- 1. Function to display usage info ---
usage() {
    echo "Usage: $0 [filename]"
    echo "This script modifies permissions and ownership of a specified file."
}

# --- 2. Check if a file name is provided or ask for it ---
if [ -z "$1" ]; then
    echo "Enter file name:"
    read FILE
else
    FILE=$1
fi

# --- 3. Check if the file exists ---
if [ ! -e "$FILE" ]; then
    echo "Error: File '$FILE' does not exist."
    usage
    exit 1
fi

# --- 4. Display current status ---
echo "--- Current Status of $FILE ---"
ls -l "$FILE"
echo "------------------------------"
```

```
  GNU nano 7.2                                                    mod-perm.sh
# --- 5. Get desired permission from the user ---
echo "Enter desired permission (e.g., 755 for scripts, 644 for text):"
read PERM

# --- 6. Change file permissions ---
chmod "$PERM" "$FILE"
if [ $? -eq 0 ]; then
    echo "Successfully updated permissions to $PERM."
else
    echo "Failed to update permissions. Check if you have rights."
fi

# --- 7. Display new permissions ---
echo "Current permissions: $(ls -l "$FILE" | awk '{print $1}')"

# --- 8. Get ownership change input ---
echo "Would you like to change the owner? (y/n):"
read ANS

if [[ "$ANS" == "y" || "$ANS" == "Y" ]]; then
    echo "Enter new owner name:"
    read OWNER
    # --- 9. Change ownership ---
    sudo chown "$OWNER" "$FILE"
```

```
    # --- 10. Display final ownership and permission ---
    echo "--- Final File Status ---"
    ls -l "$FILE"
else
    echo "Ownership change skipped."
    echo "--- Final File Status ---"
    ls -l "$FILE"
fi
```

Output:

```
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ nano mod-perm.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ chmod +x mod-perm.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$ ./mod-perm.sh
Enter file name:
test.txt
--- Current Status of test.txt ---
-rwxr-xr-x 1 rvu rvu 0 Mar 10 14:18 test.txt
------------------------------
Enter desired permission (e.g., 755 for scripts, 644 for text):
644
Successfully updated permissions to 644.
Current permissions: -rw-r--r--
Would you like to change the owner? (y/n):
y
Enter new owner name:
rvu
[sudo] password for rvu:
--- Final File Status ---
-rw-r--r-- 1 rvu rvu 0 Mar 10 14:18 test.txt
(base) rvu@rvu-OptiPlex-SFF-7020:~/Desktop/SaiCJ-OS/Lab-5$
```